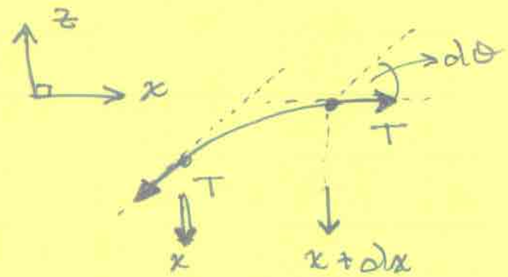
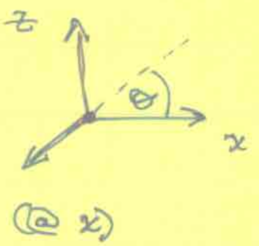
Elastic Membrane

First, consider a uniform string under tension.



◦ tension T tends to straighten the string



$\tan\theta = \frac{dz}{dx}$

$$\Rightarrow \bar{F}(x) = -T(\cos\theta\,\hat{x} + \sin\theta\,\hat{z})$$

$$\bar{F}(x+dx) = +T[\cos(\theta+d\theta)\,\hat{x} + \sin(\theta+d\theta)\,\hat{z}]$$

$$\Rightarrow \text{net force} \simeq T[-\sin\theta\, d\theta\,\hat{x} + \cos\theta\, d\theta\,\hat{z}]$$

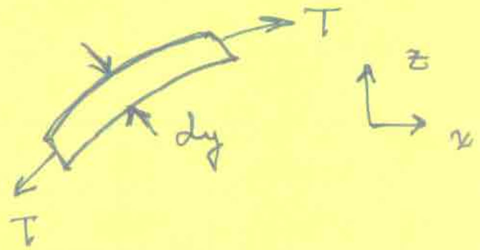
For small θ and $d\theta$: net force $\simeq \underline{\underline{T\,d\theta\,\hat{z}}}$

$$d\tan\theta = \cancel{\sec^2\theta\, d\theta} \simeq d\theta = dx\,\frac{d^2z}{dx^2}$$

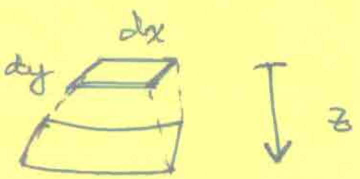
$$\Rightarrow \text{force on element of length } dx = T\,dx\,\frac{d^2z}{dx^2}\,\hat{z}$$

Note: z'' carries the sign. In the above example, $z'' < 0$, and the force restores the string to horizontal.

If we generalize from a string to a band of width dy and negligible thickness, the tension T becomes a tension per unit length in the y direction.



Now, further generalize to a 2D membrane. If we apply a pressure P in the z direction, the membrane is deflected from its equilibrium in the x-y plane.



From stretching in the x direction:

$$\text{force}_z = T\,dy\,\frac{\partial^2 z}{\partial x^2}\,dx \;.$$

In the y direction:

$$\text{force}, z = T\,dx\,\frac{\partial^2 z}{\partial y^2}\,dy \;.$$

This assumes that $\underline{T \text{ is isotropic}}$.

So, the net elastic restoring force in the $+z$ direction is

$$T\,dx\,dy\left(\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2}\right) = T\,dx\,dy\,\nabla_\perp^2 z\,.$$

In equilibrium, this must balance the pressure P.

$$\Rightarrow \quad \frac{P}{T} = \nabla_\perp^2 z\,.$$

Let $S = -z$ so that $S > 0$ is a depression

$$\Rightarrow \quad \boxed{\nabla_\perp^2 S = -P/T}\,.$$

In general, the membrane would be oscillatory (subject to boundary conditions). If $\underline{\underline{\sigma}}$ is the <u>surface</u> <u>mass density</u>, the equation of motion is

$$T\nabla_\perp^2 S = \sigma\frac{\partial^2 S}{\partial t^2} \qquad \text{(2D wave equation)}.$$

Consider a δ-function pressure in cylindrical coordinates:

$$\nabla_\perp^2 S = \frac{1}{r}\frac{d}{dr}\left(r\frac{dS}{dr}\right) = -\frac{P}{T} .$$

For $r > 0$, $P = 0$

$$\Rightarrow \quad S(r) = a \ln r + b .$$

$$\int \nabla_\perp^2 S \, dA = -\frac{w}{T} ,$$

where w is the weight associated with P $\left(w = \int dA\, P\right)$.

From Gauss' theorem,

$$\int dA\, \nabla_\perp^2 S = \oint dl\, \hat{n}\cdot\overline{\nabla}_\perp S$$

$$\overline{\nabla}_\perp S = \hat{r}\,\frac{dS}{dr} = \hat{r}\cdot\frac{a}{r}$$

$$\Rightarrow \quad 2\pi R \cdot \frac{a}{R} = -\frac{w}{T} \quad \Rightarrow \quad a = -\frac{w}{2\pi T}$$

$$\Rightarrow \quad S = -\frac{w}{2\pi T}\ln r + b .$$

Suppose we have a second boundary condition such that $S(r=R) = 0$ (fixed ring boundary).

$$\Rightarrow \quad 0 = -\frac{\omega}{2\pi T} \ln R + b$$

$$\Rightarrow \quad b = \frac{\omega}{2\pi T} \ln R$$

$$\Rightarrow \boxed{S = \frac{\omega}{2\pi T} \ln\left(\frac{R}{r}\right)} .$$

Note that the equation $\nabla_\perp^2 S = -\frac{P}{T}$ allows us to use the superposition principle:

$$\nabla_\perp^2 S = -\frac{1}{T} \sum_i P_i \quad ; \quad P_i = \omega_i \delta_i$$

$$\Rightarrow \quad \int dA \nabla_\perp^2 S = -\frac{1}{T} \sum_i \omega_i = \sum_i \int dA \nabla_\perp^2 S_i .$$

So, to linear order, we can superpose deflection fields.